Working with other visualizations and properties Activity Book

Create Visualizations with Qlik Sense

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Activity 1: Working with maps

Summary:

- Explore and work with Qlik Sense visualizations.
- Use the Insight Advisor and the Chart suggestions wizard to generate new visualizations in the app.
- Create new visualizations using the Assets panel.

Qlik Sense app: ABC Sales Analysis v2

Challenge

Analyze sales based on regions where customers are located (countries and cities) using a new sheet in the **ABC Sales Analysis v2** app.

Create a **Regional sales** sheet in the app where users should be able to:

- Identify where the customers are located and see the sales in each region (country and city). Use a map to analyze this:
  - Color the countries by Sales
  - Identify the cities with shapes (triangles) where the size and color of the shape represent the *Sales* measure
  - Show *Sales*, *Num Orders*, and *Margin%* information when hovering over a city
  - Display the country flag for each country and city in the tooltip*
  - Display the cities (city layer) only when the zoom level is 4x zoom or more
  - Display a dynamic subtitle to show the number of cities and customers
- Analyze the exact figures of *Sales*, *Num Orders*, and *Num Customers* by each country and city
  - Use a tabular representation and style text and headers
  - Highlight the rows when hovering over them

Additionally, users should be able to filter and make selections by *Division, Country, City*, and *Customer* using a filter pane on this sheet.

*Using flag images provided online in an expression. The images are located in *https://qt-lss.s3.amazonaws.com/Curriculum/Country-Flags/* folder. Each flag has the name of the country and a png extension.

Hints

- Use the following visualizations in the new sheet:
  - *Sales* map with layers: *Country* (**Area layer**) and *City* (**Point layer**)
  - *Sales, Num Orders*, and *Num Customers* by month (**Pivot table**)
- In the map:
  - Use the **Options > Layer display** section in the properties panel of each layer
  - Add a subtitle using text and count() functions; e.g. to display the number of cities:

```
count(distinct City) & ' cities'
```

Step-by-step guide

1. In the **ABC Sales Analysis v2** app, create a new sheet, *Regional Sales*, and open it in edit mode.

   > **Note**: You may need to use the **Advanced options** toggle to access the Assets panel in Edit mode.

2. Drag the *Customer* field onto the sheet with the *Shift* key pressed to create a filter pane. Then locate it on the left of the sheet.

Sales Map

3. From the **Assets** panel, add a **Map** onto the sheet next to the *Customer* filter and resize it to make it bigger.
4. In the properties panel, expand the **Map settings** and review the options for **Base map**, **Projection**, **Map language**, **Units**, and **Selection method**. Then select *Pale* for **Base map** and *Adaptive* projection.
5. Under **Appearance > General**, add a title *Sales map*.
6. Under **Layers**, click **Add layer** and select **Area layer**. Then under **Dimensions**, select **Add** and choose *Country*.
7. For this new layer, under **Colors**, set the **Colors** to **Custom** and select **By measure** in the drop-down menu. Then choose the *Sales* measure and the **Sequential gradient** for the color scheme.
8. Under **Options** > **Tooltip**, select **Custom**. Then under **Images**, select *URL* as **Type** and enter the following expressions as **URL**:

   ```
   ='https://qt-lss.s3.amazonaws.com/Curriculum/Country-
   Flags/'&Country&'.png'
   ```

   > **Note**: In Cloud editions of Qlik Sense external URLs and content access must be enabled in the Management Console, under **Content Security Policy**.

9. Click on the **Country Area layer** (at the top of the properties panel) to go back to the **Sales map** properties. Then click **Add layer** again, and choose a **Point layer** this time.
10. For this new point layer, under **Dimensions**, click **Add** and select *City*. Then under **Size & Shape**, select **Size by** *Sales* and *Triangle* as **Shape**. Notice you can also use a custom image instead of a shape.
11. Under **Colors**, make sure that the **Colors** are set to **Custom** and select **By measure** in the drop down menu. Choose *Sales* for measure and for the **Outline color** select a bright red color from the palette.
12. Under **Options > General**, turn **Show labels** on. Then, under **Options > Tooltip**, add two measures: *Num Orders* and *Margin%*. You can also add in this layer the URL Images for the country flags, as in step 8.

> **Note**: Under the **Options > Layer display** properties menu of each layer, a calculation condition can be specified to hide or show the layer. In the **Options > General** properties, you can also select **Disable selections** to avoid selections being made at this layer.

13. Under **Options > Layer display** (for the **City Point layer** properties panel), set the **Show in zoom levels** property to **Custom**, and use the scroll to initial zoom to **4x zoom**.
14. Click on the **City Point layer** (at the top of the properties panel) to go back to the **Sales map** properties.
15. Under **Appearance > General**, add the following 3 expression as the **Subtitle**:

```
=count(distinct Customer) & ' customers in ' &count(distinct City) & '
cities'
```

16. Notice there is an info icon on the map. Click on the info icon to see the warning message. The data quality of the *City* field in this data set might not be good enough for Data manager to identify all the city points in the map. In a real scenario, you need to consider these messages, as data that is not located will be missing from the map. In these cases, it is always recommended to support the visualizations with a table or other visualizations where you can investigate those details.

Pivot table

17. Add a **Pivot table** on the sheet. Use *Country* and *City* as dimension **Rows**, using the properties panel.
18. Add *Sales* and *Num Orders* as **Measures**. Then, format *Sales* using the **Money** format, disabling the Master measure format, as data that is not located will be missing from the map.
19. Add a third measure using the following 3 expression:

```
count(distinct CustomerID)
```

20. Label the new measure as *Num Customers*.
21. In the properties panel, under **Add-ons > Data handling**, clear **Include zero values**.
22. Under **Appearance > Presentation**, clear **Indent rows**. Then, select **Styling > Chart** and apply the following changes:
    - Set the **Header font size** to *14* and use the dark green for the color.
    - Set the **Content font size** to *14*.
    - Set **Highlight rows on hover** to **on** and select yellow as the color for the **Row hover color**.

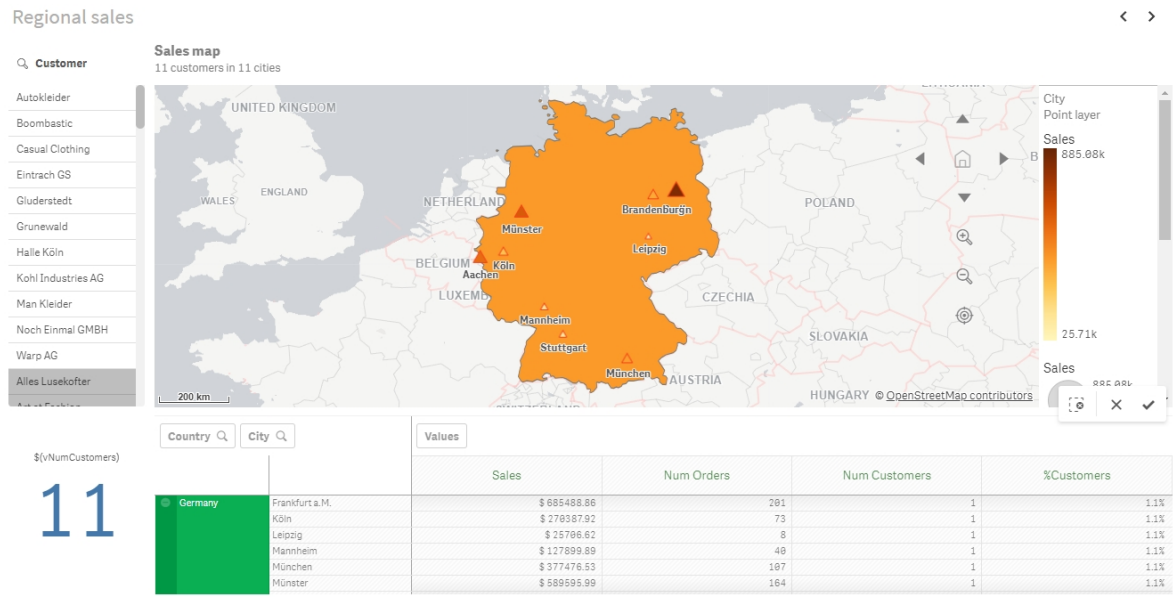
Analyze regional sales

23. In edit mode, activate the **Navigation controls** in the map from the properties panel, under **Appearance > Presentation,** set **Navigation** to **On** and **Location finder** to **On**.
24. In analysis mode, use the **Navigation controls** on the map to zoom in and review Europe, then select the country with the higher sales using the map and compare your results with the solution image. See how the selection affects the map. Notice when the city information is displayed in the map.

25. Use the Circle selection to see how you can select customers and cities within a 250km radius from *Berlin*.
26. Clear selections and use the location finder (**Find my location** icon) to locate your current location in the map. Then zoom out to see the whole country. Note that if you are using a virtual platform, your location might display as the location of the virtual server.

> **Note**: Using the map exploration menu, you can: toggle layers, define size objects, change base map, change map language, etc.

Solution



Challenge 2: Working with extension bundles

Summary:

- Explore and work with the Qlik Sense extensions.
- Use visualizations and objects in the Dashboard and Visualization extension bundles.
- Configure extension properties using the properties panel.

Qlik Sense app: ABC Sales Analysis v2

Challenge

Create a sheet in the **ABC Sales Analysis v2** called **Extensions** and add the following objects:

- The **Word Cloud chart** extension with *Product category* as the dimension and *Sales* as the measure - use the **Appearance** section in the properties to customize this extension as follows:
  - Change the orientation of the text to horizontal, setting **Start Angle** to *0*
  - Change the **Font Min Size** to *12* and the **Font Max Size** to *40*
  - Change the font color to set a color range using **Enable Custom Range**
- The **Radar chart** extension with *Quarter* and *Product category* as the dimensions and the *Sales* as the master measure.
- The **Trellis container** extension with *Division* as the dimension and the *Sales trend* as the master visualization (this line chart is available as a master visualization in the app).
- The **Straight table** extension with *Division*, *Country*, *Product category*, *Product* and *Customer* as the dimensions and *Sales*, *Margin*, *Discount*, *Num Customers* and *Num Orders* as the measures - use the **Appearance > Presentation** section to enable the **Chart exploration** panel.

Add the following extensions from the **Qlik Dashboard** bundle:

- **Date picker** - select from the *OrderDate* field, a range of dates using the calendar format.
- **Animator** - select the *Week* field to animate all the visualizations in the sheet by this field.
- **NL Insights** - select *Sales* as the measure and *Division* and *CategoryName* as the dimensions.

Qlik Sense Enterprise / Cloud editions of Qlik Sense: These extensions have already been included into your environment and they are available under the **Custom objects** section in the Assets panel. Select **Qlik Dashboard bundle** or **Qlik Visualization bundle** to see all the related objects.

Qlik Sense Desktop: The Qlik Extension bundles are installed during the QSD installation process if you selected that option. If you did not include the bundles when you installed QSD, you can still import these extensions:

1. Download the extension bundle files (available in the Student files) on your computer.
2. Using the File Explorer, navigate to **C:\Users\<username>\Documents\Qlik\Sense\Extensions** folder.

3. Unzip the extension bundle files you downloaded, extracting them in the **Extensions** folder. The extensions are now available in Qlik Sense Desktop.

Then make selections in the **Date Picker** extension and the other visualizations to analyze the data to answer the questions below:

- How were the sales in each division for the period: *4th Jan 2023 - 31th March 2023*? (Use the **Date picker** to select the dates)
- What are the top 3 selling product categories for the same period? (Hover over the **Radar chart** to see the sales values)
- In the *Baby Clothes* and *Swimwear* categories, how were the sales in the different divisions in this period? (You can make selections in the **Word cloud chart**)
- What is the full trend of each division over all the years for *Swimwear* products? What quarter is the strongest overall for this category? (Use the **Word cloud chart** to make the product category selection, use the **Radar chart** to see what quarter is the strongest over all regions, then use the **Trellis container** to review the trend and select the latest year in one of the trends to see the details per month of this year.)

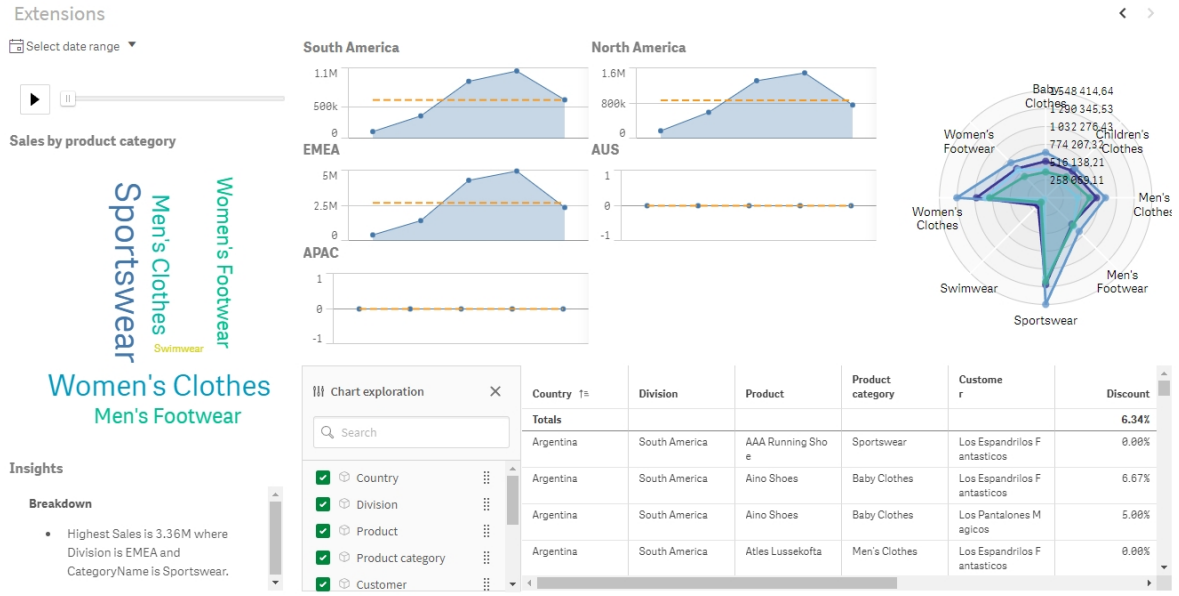
To finish this challenge, play the animation in the sheet to see how the product categories trend over the weeks. Then discuss other interesting uses for the **Animator** extension. Remember to clear the selections once you finish working with this extension.

Note: The objective of this **challenge** is to test whether you feel comfortable with the concepts that have been discussed so far in your class. There is no **Step-by-Step** section in this activity, so you must only rely on the **Hints** below to understand the tasks you have to execute.
As you are still new to many of these concepts, please don't hesitate to reach out for more guidance (help file, instructor, Qlik Community).

Hints

- From the **Assets** panel, under **g Custom objects**, you can add all the extensions to your sheet:
  - Under the **Qlik Visualization bundle**, you can add the **Radar chart**, **Word Cloud Chart**, **Trellis Container** and the **Straight table**.
  - Under the **Qlik Dashboard bundle**, you can add the **Date picker**, **Animator** and the **NL Insights**.
- Use the properties panels to configure extensions as you do with other standard charts.

Solution


Extensions
Select date range
Sales by product category
Sportswear
Men's Clothes
Women's Footwear
Swimwear
Women's Clothes
Men's Footwear
Insights
Breakdown
Highest Sales is 3.36M where Division is EMEA and CategoryName is Sportswear.
South America
North America
EMEA
AUS
APAC
Chart exploration
Search
Country
Division
Product
Product category
Customer
Country
Division
Product
Product category
Customer
Discount
Totals
6.34%
Argentina
South America
AAA Running Shoe
Sportswear
Los Espandrilos Fantasticos
0.00%
Argentina
South America
Aino Shoes
Baby Clothes
Los Espandrilos Fantasticos
6.67%
Argentina
South America
Aino Shoes
Baby Clothes
Los Pantalones Magicos
5.00%
Argentina
South America
Atles Lussekofta
Men's Clothes
Los Espandrilos Fantasticos
0.00%
Women's Footwear
Children's Clothes
Men's Clothes
Men's Footwear
Sportswear
Swimwear
Women's Clothes

Activity 3: Using properties to define visual aspects

Summary:

- Work with the visualizations properties.
- Define specific colors for values and segments.
- Use sorting and shape features in objects.
- Apply background options.

Qlik Sense app: ABC Sales Analysis v2

Challenge

In the **ABC Sales Analysis v2** app, provide more information and context, and give more insights by including visual aspects in these sheets:

ABC Dashboard

Use color to highlight the comparison between current and last year sales:

- Highlight the *Sales CYTD - LYTD* bar chart by coloring the bars red if they are below 0 or green if they are above 0.
- Use the *Sales CYTD - LYTD* master measure to set the segment colors.

Product overview

Use a silent legend based on the *Product category* dimension, and apply it to the relevant visualizations in the sheet.

- Set the **Value colors** for the values of the *Product category* master dimension. Use a different color for each value, but choose similar colors for these category items: Footwear (all values containing footwear), Clothes (all values containing clothes), and Sports (Swimwear and Sportswear).
- Apply this color library based on *Product category* to the visualizations in the sheet using this dimension: donut, treemap, and bar chart.

Display each product photo in the tooltip in both bar chart and treemap. The product png images can be found in this location:

https://qt-lss.s3.amazonaws.com/Curriculum/Product-Photos/

The corresponding png file for each product is labeled the same as the product name. If the product name contains spaces, the image file will have the spaces replaced with a "+" symbol. Example: For the product called *Bow tie*, the image file is called *Bow+tie.png*

Customer Sales

Use the **Customer sales details** table to highlight discount rates above the 7.2% threshold with color and compare sales for the current year versus previous year with shape and color indicators, and the picture of each product per row. Include also a chart that allow the user to see the weight of each region in Sales by Customer Type.

- Values below that threshold must display on a green background, values above that threshold must be displayed on a red background.

- Add an indicator in this table for the *Sales CYTD - LYTD* column. The indicator will show up and down arrows (▲ and ▼) in green and red colors, respectively, to represent the *Sales CYTD - LYTD* variance.
- Use the photo of each product per row in the table object.
- Add a master visualization to show sales by each region in the pie chart.

Sales analysis

Include information for the *Avg deal size** key measure and guide users on the app selections using buttons:

- Add an *Avg deal size** KPI and use color and shapes to highlight the value in red with a warning symbol when it is under 3.5k.

  Link the KPI with the **Customer orders** sheet, so by clicking on this KPI, the user can navigate to this sheet.

  Use a title ("*Avg Deal Size*") and footnote ("*Sales / Order volume*").

  Display information so users can see the info icon when hovering over the object.
- Sort the *Customer* values in the combo chart and the pivot table based on the *Avg Deal Size** in descending order.
- Create two buttons to allow guided selections:
  - The **Americas** button to toggle selections in the *Division* field, selecting or clearing selections for the Americas regions (*North America* and *South America*). You can use the * wildcard: *(*America)*.
  - The **Current Year H1** button to select the current year and the H1 period. In this case use a search: *(>=01/01/2023<=06/30/2023)*.

*The *Avg deal size* is calculated as the sales amount divided by the number of orders.

- Insert a custom background to the pivot table using a color and an image.

Hints

- For the **ABC Dashboard** sheet, set **Segment colors** for the master measure called *Sales CYTD - LYTD* and use the Library colors in the chart properties when using this measure to color the chart.
- For the **Product overview** sheet,
  - Master items can be modified when a sheet is open in edit mode. From the **Assets** panel, you can select the master item you want to modify and use the edit @ button. The color for the measure or dimension can be set from the edit dialog.
  - The **Value colors** tab is available from the edit dialog of the master dimensions. The **Colors and legend** section in the **Properties** panel allows you to set the colors in a chart.
- For the **Customer Sales** sheet,
  - The **Background color expression** property defines the color of the column in a table. The **Discount** column already exists in the table but an expression is required to define the color expression. Colors can be defined with the **RGB()** function. For instance, a red color can be represented as: *RGB(170,57,57)* and a green color as: *RGB(87,149,50)*. You can modify these colors by altering the 3 parameters in the function.
  - Change colors dynamically, based on a comparison, by using an IF statement. Using a condition as the first parameter determines what expression is evaluated (the first

expression is returned if the condition is met and the second one is returned if it is not met). The syntax is the following: *IF(condition, expression 1, expression 2).*

- Use an indicator representation in the *Sales CYTD - LYTD* column. Using a limit (0) to show positive numbers in green with the ▲ symbol, or negative numbers in red with the ▼ symbol. Other Unicode character sets can be found in the following Qlik community post.
- The **Representation** option inside each column of the table allows us to show the values as images.
- Master visualizations can be shown when hover over the charts, using the **Tooltip** option under the Appearance section.

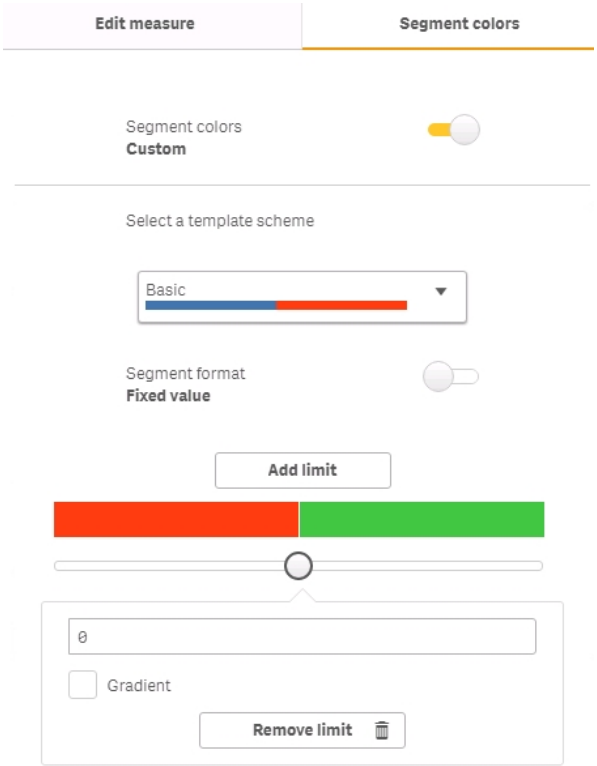
- For the **Sales Analysis** sheet,
  - The *Avg deal size* has not been created as a master measure. It is possible to add a measure in a visualization using the expression editor. Remember to add a label for the expression.
  - Use **Conditional colors** and define the segments, adding a limit, to customize color and shape.
  - KPIs can be linked with existing sheets in the app using the **Link to sheet** feature in the properties panel.
  - In the properties panel, use the **Sorting** option to sort the dimension by the *Sales* expression.
  - Use two buttons with a toggle field selection action in the corresponding fields: *Division* and *OrderDate*.
  - **Background** options can be found under **Presentation > Styling**, both in charts and sheets.

Step-by-step guide

1. Open the **ABC Sales Analysis v2** app and navigate to the **Analyze > Sheet** area.

ABC Dashboard sheet

2. Go to the **ABC Dashboard** sheet and turn the edit mode on.
3. Using the Assets panel, select **Master items > Measures > Sales CYTD - LYTD**. Then click **Edit@** and **OK**.
4. Select the **Segment colors** tab and set the **Segment colors** to **Custom**. Then set the template scheme to **Basic**, the **Segment format** to **Fix value**, and set the default limit to *0*.
5. Press the first segment and set the color to red. Then for the second segment, set the color to green and save the settings.



6. Select the **Sales CYTD - LYTD** bar chart in the sheet to review the chart properties. Under **Appearance > Colors and Legend**, set the **Library colors** to **On** and clear the **Reverse colors**.
7. In analysis mode, you can make selections in this chart to review that the color result in the bars is the expected one, negative values are red bars and positive values are green bars.

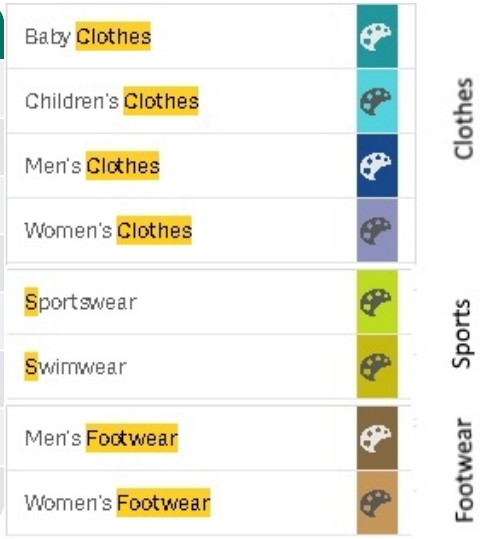
Product overview

8. Clear selections and open the **Product Overview** sheet in **Edit mode**.
9. From the **Assets** panel, under **Master items > Dimensions**, select the *Product category* master dimension and click the edit @ button in the information pop-up.
10. Select the **Value colors** tab from the edit dialog of the master dimension. Then apply different colors to each value as follows:
    - Click on the corresponding color for that value
    - Use the color palette to select a different color

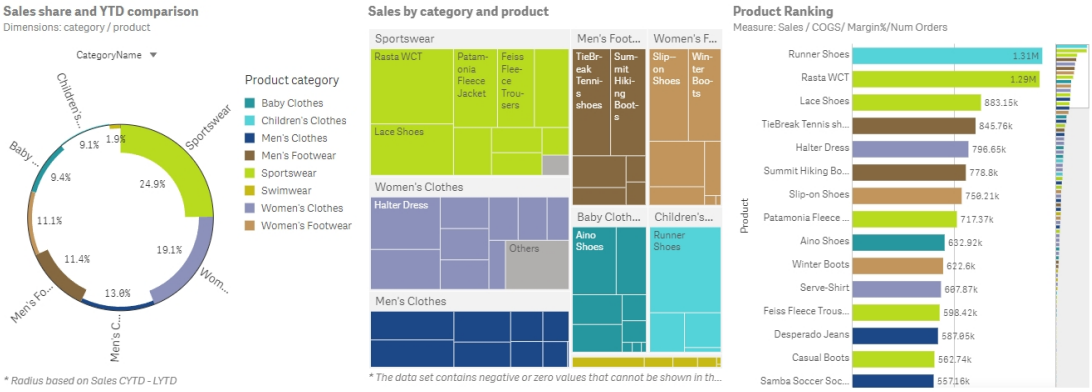
    You can also apply a default color for others and null values.

11. Modify the bar chart, pie chart, and treemap properties: under **Appearance > Colors and Legends**, set **Colors** to **Custom By Dimension**, select *Product category*, and make sure **Library colors** is **on**.

Product category	Color Code
Baby Clothes	#27979f
Children's Clothes	#54d3d9
Men's Clothes	#1c4888
Men's Footwear	#866840
Sportswear	#b8db20
Swimwear	#c7ba17
Women's Clothes	#8c91bc
Women's Footwear	#c1955c



You can "group" the different categories using similar colors.



12. In the bar chart and treemap properties, go to **Appearance > Tooltip** and set it to **Custom**. Then under *Images*, select *URL* as **Type** and enter the following expression as **URL**:

```
='https://qt-lss.s3.amazonaws.com/Curriculum/Product-Photos/' &
Replace(Product,' ','+') & '.png'
```

Customer Sales

13. Navigate to the **Customer Sales** sheet and select the **Customer sales details** table.
14. In the **Properties** panel, under the **Discount** column, set the **Background color expression** using the following expression:

```
If (Avg(Discount)>=0.072, RGB(170,57,57), RGB(87,149,50))
```

15. For the *Sales CYTD - LYTD* column in the table, set the **Representation** to **Indicator**. Then add a limit, set it to *0*, and set the color and shapes to the limit. Use the ▲ and ▼ icons, to represent negative or positive values.
16. Set the **Indication Position** to **Right** and check the **Apply color to value** option.
17. In analysis mode, sort the table by *Customer* by double clicking on the first column label:

Customer	Region	Disco...	Margi...	Sales ($k)	Sales CYTD - LYTD	Sales trend	Last order placed
Alles Lusekofter	EMEA	6.09%	19.38%	$44.06	-$ 10824.38 ▼		10/18/2022
Art et Fashion	EMEA	5.50%	21.90%	$216.82	$ 31784.23 ▲		05/17/2023
Aujourd´hui	EMEA	6.14%	22.28%	$85.05	$ 6303.35 ▲		06/02/2023
Autokleider	EMEA	6.92%	20.81%	$108.97	-$ 24365.40 ▼		06/07/2023
Belgium Black Jeans	EMEA	5.54%	21.84%	$131.48	$ 2360.23 ▲		06/14/2023
Big Foot Shoes	Americas	8.09%	19.66%	$137.98	-$ 4346.81 ▼		06/17/2023

> Note: Once you sort the table or resize columns in analysis mode, the properties of the table change. If you want to edit the properties of the table again (in edit mode), you need to **Apply** (or **Discard**) the changes you applied in analysis mode.

18. Include *Product* as a new dimension. For this column, set the **Representation** to **Image**, and **Image setting** as **Add URL**. In the **Image URL** enter the following expression:

```
='https://qt-lss.s3.amazonaws.com/Curriculum/Product-Photos/' &
Replace(Product,' ','+') & '.png'
```

19. In the pie chart properties panel, under **Appearance > Tooltip**, set Tooltip to **Custom**. Select **Add chart** and choose *Sales share*. the Then, on analysis mode, hove over the chart to check that the weight of Americas is higher in Silver sales than in Gold sales.

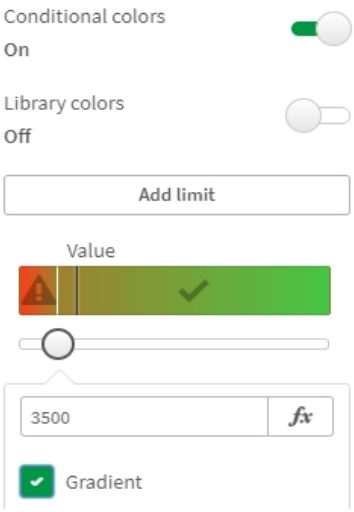
Sales Analysis

20. Open the **Sales Analysis** sheet in edit mode, reduce the size of the container, and add a **KPI** in a free space on the sheet.
21. Use the properties panel to add the measure using the following expression:

```
[Sales]/[Num Orders]
```

.
Then label the measure as *Avg deal size* and click **Add new**, under **Master item**, to add it as master measure in the app. To save the new master measure, click **Create**.

22. In the KPI properties panel, under **Appearance > Color**, set the **Conditional colors** to **On** and the **Library colors** to **Off**.
23. Add a limit of *3500*. For the left segment, set the color to red and choose the **è** warning symbol. For the right segment, set the color to green and choose the **m** tick symbol. Check the **Gradient** box.



24. Under **Appearance > Presentation**, set **Link to sheet** to **On**. Then use the drop-down and select the **Customer Orders** sheet.
25. Under **Appearance > General**, set **Title** to *Avg deal size** and **Footnote** to **Sales / Order volume*. Then set **Show details** to **Show**.
26. Under **Appearance > Presentation**, clear **Show first KPI title**.
27. Let's sort the customers based on this measure. In the existing container, access the combo chart properties. Then, in the properties panel of the combo chart, under **Sorting**, make sure *Customer* is the first field on the list.
28. Set the **Sorting** to **Custom**, **Sort by expression** for this dimension using the **Expression**:

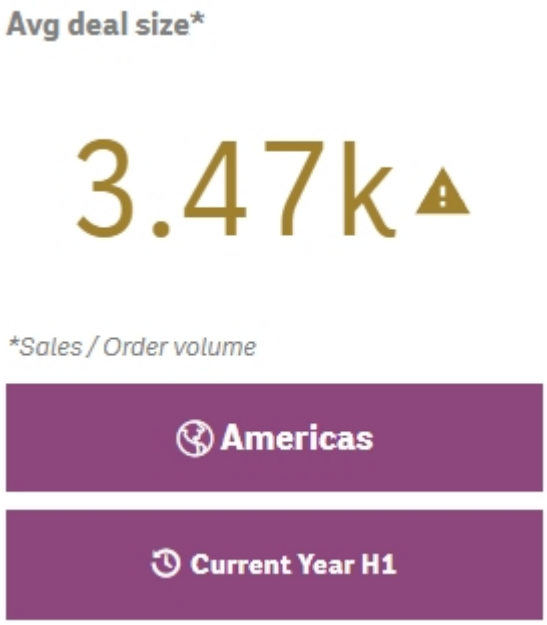
    ```
    [Avg deal size]
    ```

    .
    and select **Descending**. Note that you can add a subtitle in both charts to explain the sort order to the end users.
29. Repeat the same steps for the pivot table.
30. From the Assets panel, under **Charts**, drag and drop a **Button** into the sheet. Then select it to configure the settings in the properties panel as follows:
    a. Add an action, selecting **Toggle field selection** as **Action**, label this action as *Americas*, use *Division* as **Field**, and the following expression as **Value**:

    ```
    '*America'
    ```

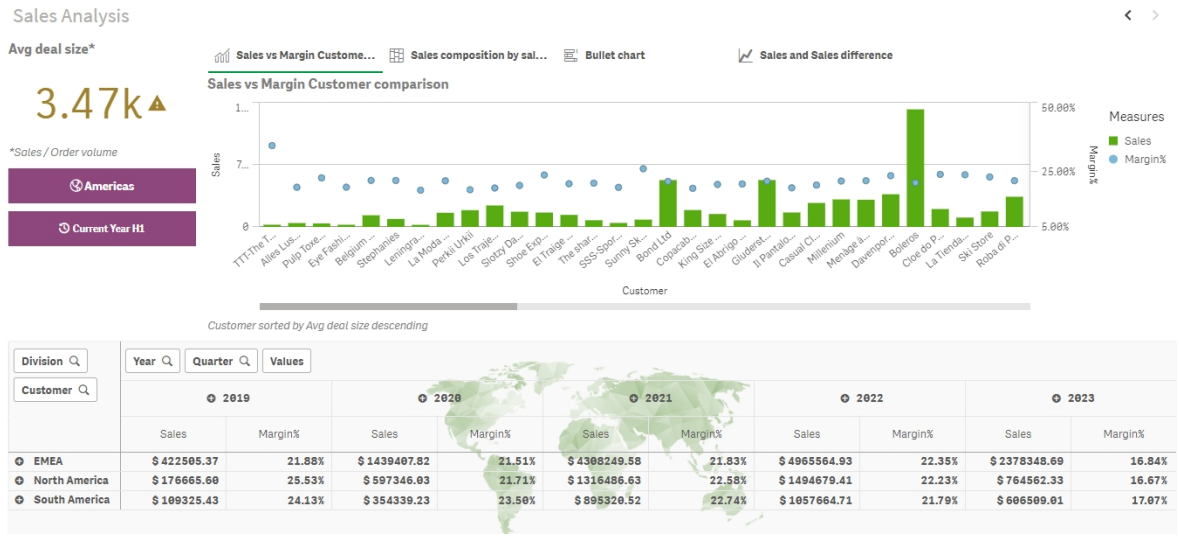
    b. Under **Appearance > General**, change the **Label** to Americas.
    c. Under **Appearance > Background**, change the color background, then under the **Icon** property, set **Show Icon** to *On* and add a *Map* icon from the drop-down list.

    Repeat this step again to add a new button, this time using the *OrderDate* field as **Field** in the action and the *(>=01/01/2023<=06/30/2023)* as the **Value**. Label this button **Current Year H1** and apply the *History* icon.



31. In analysis mode, select *EMEA, 2023*, using the pivot table (to select *EMEA*) and the *Sales and Sales difference* chart in the container (to select the year). Notice how the KPI number changes. Clear selections, click on the **Americas** button and the **Current Year H1** button. Then click the **Americas** button again to clear the selections in the *Division* field.
32. Hover over the KPI and click the **]** info button. An info dialog appears with the details about the expression used for this calculation.
33. Close the **Details** panel and click on the KPI once to see the details about the sheet that is linked. Then, click again to navigate to the **Customer orders** sheet.
34. Come back to the **Sales Analysis** sheet and click on the pivot table at the bottom of the sheet. Go to **Appearance > Presentation > Styling**.
35. Under General section, choose a **Single color** as the **Background color** and type: *#f9f9f9*.
36. In the **Background image** option, select **Media library**. Select **Upload media** to upload the *WorldMap.png* file, that can be found in the StudentFiles folder of this module. When the image is already loaded, select **In app** and choose the file we have just uploaded.
37. Click **Insert**. Select **Always fit** and the bottom-center position.

Solution



Division
Customer
Year
Quarter
Values

	2019		2020		2021		2022		2023	
	Sales	Margin%	Sales	Margin%	Sales	Margin%	Sales	Margin%	Sales	Margin%
EMEA	$ 422505.37	21.88%	$ 1439407.82	21.51%	$ 4308249.58	21.83%	$ 4965564.93	22.35%	$ 2378348.69	16.84%
North America	$ 176665.60	25.53%	$ 597346.03	21.71%	$ 1316486.63	22.58%	$ 1494679.41	22.23%	$ 764562.33	16.67%
South America	$ 109325.43	24.13%	$ 354339.23	23.50%	$ 895320.52	22.74%	$ 1057664.71	21.79%	$ 606509.01	17.07%

Activity 4: Interacting with user behaviors

Summary:

- Apply dynamic calculations to visualizations and objects in sheets to create interactive text and actions.
- Use calculation modifiers, trend lines, and reference lines to provide context and insights.
- Include calculate conditions and show conditions to maximize user attention focus and create a guided analysis.

Qlik Sense app: ABC Sales Analysis v2

Challenge

Apply some dynamic and interactive responses in the **ABC Sales Analysis v2** app:

In the **ABC Dashboard** sheet, modify the sheet title and sheet description to provide more context:

- For the sheet title, use the basic title but include information for the selected *Year* and *Region* values. Always display all the possible or available regions and only the selected years. If there are no regions selected, list all the regions and if there are no selected years, do not display any years. Title examples:
  - With no selections, the title should read: *ABC Dashboard - Americas, Asia & Australia, EMEA*.
  - With *2022* and *2023* years and *Americas* region selected, the title should read: *ABC Dashboard 2022, 2023 - Americas*.
- For the sheet description, include a sheet description and display the following information: Last order date, Last reload date, and Today's date.

The **Product Analysis** sheet analyzes relationships between different products correlated by different measures. This sheet should be hidden if a single product is selected.

In the **Sales analysis** sheet, adjust the container properties to show the waterfall chart only when the data contains a single year:

- Apply a show condition to the waterfall chart embedded in the container.
- The condition should show the chart when one single year is selected or available based on other selections.

In the **Customer sales** sheet, show only relevant information and enable buttons based on the data selected. Additionally, classify customers dynamically in three tiers to allow advanced customer classification based on selections:

- If the selection has 5 or fewer customers, the table should be rendered with the calculated results.
- If the selection has more than 5 customers or no customers have been selected, the table should not display the calculated result, and instead, display an appropriate message.
- When the table is rendered with results and only one *Region* is selected or available, a new column with information about the customer's country and city in the following format: *Country (City)* should be shown. Otherwise, these columns should not be displayed.
- If there is only one customer, enable the **Customer orders** button, otherwise disable it.

Hints

- In order to create dynamic text, you can concatenate static text and expressions using the ampersand (**&**) symbol. Use single quotes for the static text and expressions for dynamic text. Use the **GetFieldSelections()** and **Concat()** functions to get the list of selected or possible values in a field. You can add extra parameters to functions to include separators for the values. Notice that you might need to use **Distinct** when applying the **Concat** function.
- The *Year* field is an auto-generated field and in order to use it in an expression, you will need to refer to it as *[OrderDate.autoCalendar.Year]*. Use the **Max()** function on the *OrderDate* that returns the maximum value available. This function is applicable to numeric and date fields. When applied to dates, you might need to format the numeric date value again using the Date() function. Use other date functions to represent last reload and today's date: ReloadTime() and Today().
- A Show condition can be set to each object in a container object. In the container properties, under **Content**, select the corresponding visualization and use the **Show Condition** input box to apply the condition.
- A calculation condition in the table can be set using **Add-ons** in the properties panel, under the **Data handling** section. Each table column can also be hidden or displayed based on a condition using the **Show column if** property, under the **Data** section in the properties panel.
- A button can be disabled or enabled based on a calculation condition. In the **Enable chart** section of the button properties, set the **Condition to enable chart** option on.

Step-by-step guide

1. In the **ABC Sales Analysis v2** app, open the **ABC Dashboard** sheet in edit mode.

ABC Dashboard

2. In the **Sheet** properties panel, add the following **Title Expression**:

```
='ABC Dashboard '& GetFieldSelections([OrderDate.autoCalendar.Year],',
')&' - '& Concat(distinct Region,', ')
```

3. In the **Sheet** properties panel, add the following **Description**:

```
='Last order date '& Date(Max(OrderDate),'MM/DD/YYYY')&chr(10)&
'Reload date '& Date(ReloadTime(),'MM/DD/YYYY')&chr(10)&
'Today is '& Date(Today(2),'MM/DD/YYYY')
```

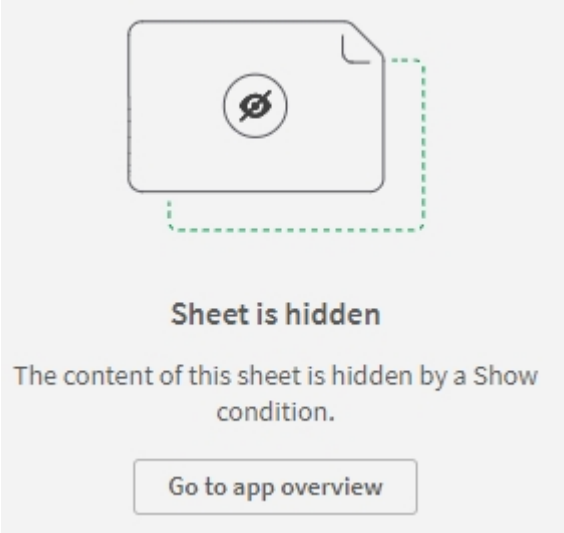
4. In analysis mode, select *EMEA* using the pie chart, then select *2022* and *2023*, using one of the line charts. Note that the dynamic title changes. Then go to the **Sheets** area, hover over the **ABC Dashboard** sheet, and click on the] info icon to see the description.

Product analysis

5. Open the **Product analysis** sheet in edit mode.
6. In the **Sheet** properties, under **Show condition**, enter the following expression:

```
Count(Distinct ProductID)>1
```

7. In analysis mode, select one single product using the scatter plot in this sheet and check the results. You can navigate to the app overview and see that the sheet is not available anymore.



Sales analysis

8. Open the **Sales analysis** sheet in edit mode, clear selections, and select the existing container object.
9. In the properties panel, under **Content**, expand the *Sales and Margin composition* waterfall chart and under **Show condition**, type the following expression:

```
count(distinct [OrderDate.autoCalendar.Year])=1
```

10. Turn analysis mode on, and test the condition by selecting one single year. Then clear selections and turn edit mode on.

Customer sales

Table conditions

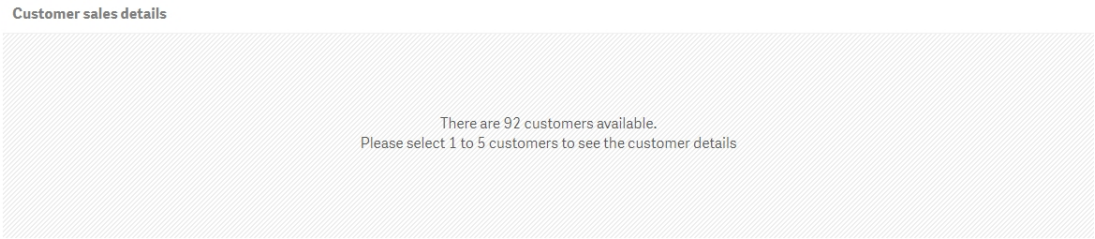
11. Navigate to the **Customer sales** sheet and edit the **Customer sales details** table.
12. Using the properties panel (under **Add-ons** > **Data handling**), add the following expression in the **Calculation condition** field:

```
Count(DISTINCT CustomerID) <= 5
```

13. Modify the **Displayed message** to use the following expression:

```
='There are '&count(Distinct CustomerID)&' customers available.
Please select 1 to 5 customers to see the customer details'
```

When there are more than 5 customers in the data set the table displays like this:



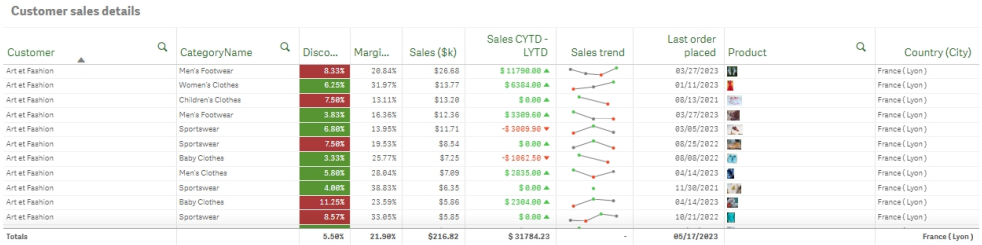
14. Using the properties panel (under **Data > Columns**):
    a. Add a new column (measure) in the table, with the label: *Country (City)*, and with the expression:

    ```
    =Country& ' ( '& City &' )'
    ```

    b. Define the following expression for the **Show column if** property:

    ```
    count(distinct Region) =1
    ```

So when there is only one region selected or available, the table displays like this:



Customer sales details

Customer	CategoryName	Disco...	Margi...	Sales ($k)	Sales CYTD - LYTD	Sales trend	Last order placed	Product	Country (City)
Art et Fashion	Men's Footwear	8.33%	28.84%	$26.68	$ 11790.00 ▲		03/27/2023		France (Lyon)
Art et Fashion	Women's Clothes	6.25%	31.97%	$13.77	$ 6384.00 ▲		01/11/2023		France (Lyon)
Art et Fashion	Children's Clothes	7.50%	13.11%	$13.20	$ 0.00 ▲		08/13/2021		France (Lyon)
Art et Fashion	Men's Footwear	3.83%	16.36%	$12.36	$ 3309.60 ▲		03/27/2023		France (Lyon)
Art et Fashion	Sportswear	6.88%	13.95%	$11.71	-$ 3009.90 ▼		03/05/2023		France (Lyon)
Art et Fashion	Sportswear	7.50%	19.53%	$8.54	$ 0.00 ▲		08/25/2022		France (Lyon)
Art et Fashion	Baby Clothes	3.33%	25.77%	$7.25	-$ 1062.50 ▼		08/08/2022		France (Lyon)
Art et Fashion	Men's Clothes	5.88%	28.04%	$7.09	$ 2835.00 ▲		04/14/2023		France (Lyon)
Art et Fashion	Sportswear	4.00%	38.83%	$6.35	$ 0.00 ▲		11/30/2021		France (Lyon)
Art et Fashion	Baby Clothes	11.25%	23.59%	$5.86	$ 2304.00 ▲		04/14/2023		France (Lyon)
Art et Fashion	Sportswear	8.57%	33.05%	$5.85	$ 0.00 ▲		10/21/2022		France (Lyon)
Totals		5.50%	21.90%	$216.82	$ 31784.23	-	05/17/2023		France (Lyon)

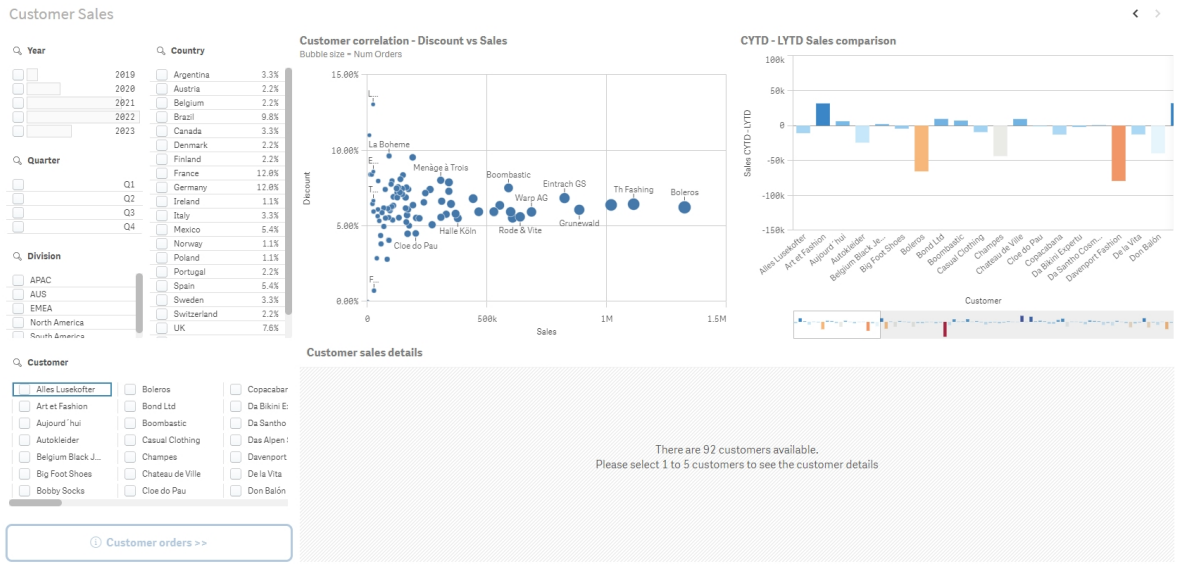
Button conditions

15. Select the button in the sheet, then under **Enable chart**, set the **Condition to enable chart** to **On** and add the following **Calculation condition**:

    ```
    count(distinct CustomerID) =1
    ```

16. In analysis mode, clear all the selections and review the disabled button and the message in the table. Then, make selections in the scatter plot to verify the calculated condition works fine. What happens when you select the 3 customers with the highest sales amounts? The table displays. And when you select only one region? A new column appears in the table displaying the country and city information for the customer.

Solution


Customer Sales
Year
2019
2020
2021
2022
2023
Quarter
Q1
Q2
Q3
Q4
Division
APAC
AUS
EMEA
North America
Country
Argentina 3.3%
Austria 2.2%
Belgium 2.2%
Brazil 9.8%
Canada 3.3%
Denmark 2.2%
Finland 2.2%
France 12.0%
Germany 12.0%
Ireland 1.1%
Italy 3.3%
Mexico 5.4%
Norway 1.1%
Poland 1.1%
Portugal 2.2%
Spain 5.4%
Sweden 3.3%
Switzerland 2.2%
UK 7.6%
Customer
Alles Lusekofter
Art et Fashion
Aujourd 'hui
Autokleider
Belgium Black J...
Big Foot Shoes
Bobby Socks
Boleros
Bond Ltd
Boombastic
Casual Clothing
Champes
Chateau de Ville
Cloe do Pau
Copacabar
Da Bikini E:
Da Santho
Das Alpen :
Davenport
De la Vita
Don Balón
Customer orders >>
Customer correlation - Discount vs Sales
Bubble size = Num Orders
Discount
15.00%
10.00%
5.00%
0.00%
0
500k
1M
1.5M
Sales
La Boheme
Menage à Trois
Boombastic
Eintrach GS
Warp AG
Th Fashing
Boleros
Grunewald
Halle Köln
Rode & Vite
Cloe do Pau
CYTD - LYTD Sales comparison
Sales CYTD - LYTD
100k
50k
0
-50k
-100k
-150k
Customer
Customer sales details
There are 92 customers available.
Please select 1 to 5 customers to see the customer details

Activity 5: Working with variables

Summary:

- Explore and work with Qlik Sense variables.
- Understand when variables are evaluated and the differences.
- Use the **Variable input** extension (part of the **Qlik Dashboard bundle**) to modify variable values.

Qlik Sense app: ABC Sales Analysis v2

Dependencies: Before working in this activity, make sure you complete Activity 1.

Challenge

In the **ABC Sales Analysis v2** app, use variables to improve the analysis and user experience in these sheets:

ABC Dashboard sheet, allow business users to dynamically change the number of customers and products included in this ranking for the two bar charts representing the top ten customers and products based on sales.

- Create a variable, *vTopNumber*, to store the number of bars to display in the charts (originally with a value of 10).
- Use the new variable in the Dimension limits of the charts to dynamically display the top customers and products specified by the variable. Also include dynamic titles in these charts to represent how many bars are being displayed by the variable.
- ***Bonus Point:*** Use the **Variable input** extension (part of the **Qlik Dashboard bundle**) to allow end users to interact with the *vTopNumber* variable definition.

In the **Regional sales** sheet, represent the number of customers and the percentage of customers per country and city. Work with variables to simplify calculations:

- Create a KPI with the total number of customers.
- In a new column in the pivot table, display the percentage of customers per country and city over the total number of customers.
- Use the following variables in the KPI and pivot table:
  - *vTotalCustomers*, with an expression that evaluates the total number of distinct customers.
  - *vNumCustomers*, with an expression (in plain text) to get the number of distinct customers. This variable will contain the expression but it does not evaluate it until the variable is used.

Hints

- You can create variables when editing a sheet, using the icon.
- A variable with an equal sign in front of the value evaluates the expression globally. This can be used to evaluate the total number of customers.
- Variables can be used in expressions and visualizations using the dollar sign expansion: *$(VariableName)*.

- Calculate the percentage of customers in each country by dividing the *vNumCustomers* variable by the *vTotalCustomers* variable.

Step-by-step guide

1. In the **ABC Sales Analysis v2** app, open the **ABC Dashboard** sheet in edit mode.

   ### ABC Dashboard

2. Click the Ò icon on the bottom left corner. The **Variables** dialog opens.
3. Click **Create new** and enter the following variable details:
   - *vTopNumber* as **Name**
   - *10* as **Definition**
4. Click **Create** to save the variable and then click **Close**. The variable is now available to use.

   Note: Variable names are case sensitive in Qlik Sense. In order to calculate and use this variable in an expression, apply the following syntax:
   `$(vTopNumber)`

5. Select the *Top 10 selling customers* bar chart. Then in the bar chart properties change the following:
   - Under **Data > Dimensions**, select the dimension and set the **Limitation** using the expression below:

     ```
     $(vTopNumber)
     ```

   - Under **Appearance > General**, set the title to this expression:

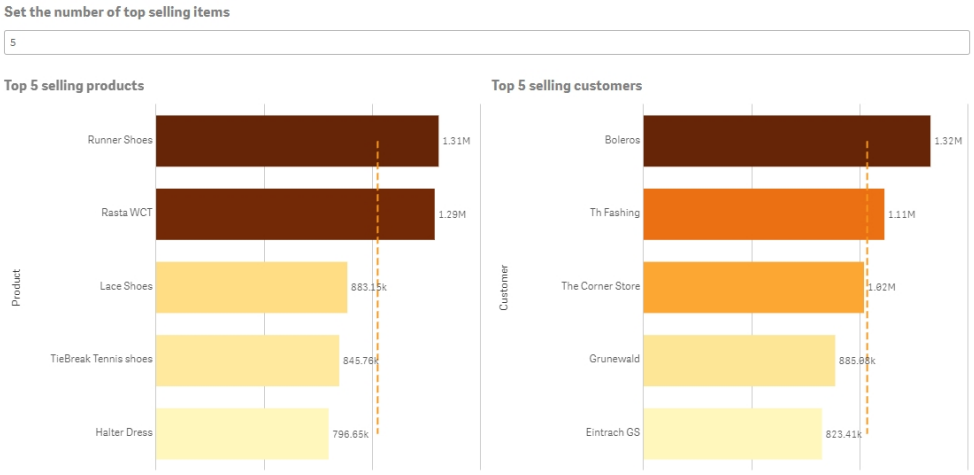
     ```
     ='Top '&$(vTopNumber)&' selling customers'
     ```

6. Repeat the previous step for the *Top 10 selling products* bar chart.
7. Change the *vTopNumber* variable definition to *5*. Then in analysis mode, review the charts.

   ### ***Bonus Point:***

   ### Use the variable input extension (Qlik Dashboard Bundle)

8. In edit mode, create some space between the text and image objects and the bar charts and drag a **Variable input** from the **Assets** panel > **Custom objects** (under **Qlik Dashboard bundle**).
9. Use the object properties, under **Appearance > General**, to add a **Title**, *Set the number of top selling items*.
10. Under **Appearance > Variable**, select the *vTopNumber* variable (and **Show as Input box**). Then stop editing the sheet and notice how users can change the variable value using this object and how the bar charts react when the value is changed.



Regional sales

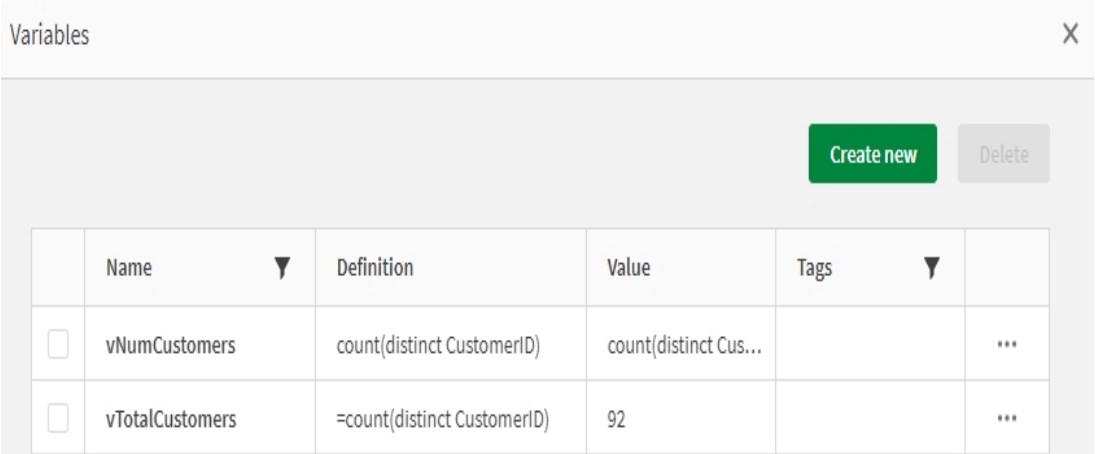
11. Open the **Regional sales** sheet in edit mode.
12. At the bottom of the **Assets** panel, open the **Variables** overview and click **Create new** to create a new variable, *vNumCustomers*.
13. Type the following expression as the variable **Definition** and save the variable by clicking on the @ icon:

```
count(distinct CustomerID)
```

14. Add a second variable, *vTotalCustomers* using the following **Definition**:

```
=count(distinct CustomerID)
```

15. Save the variable. Now there are two new variables in the app:



16. Add a new **KPI** in the sheet next to the pivot table.
17. Type the following expression as the measure for the new KPI:

```
$(vNumCustomers)
```

You can include a label as well, *Num Customers*.

> **Note:** Variable names are case sensitive in Qlik Sense. In order to calculate a variable, use the following syntax:
> `$(variableName)`

18. Select the pivot table and use the **Properties** panel to add a new measure:

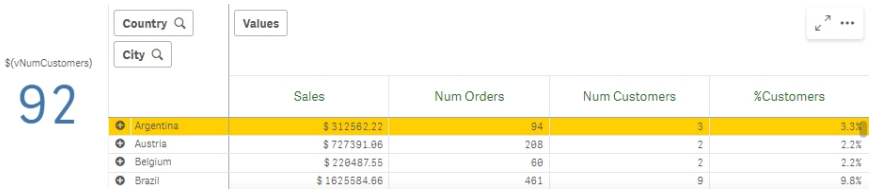
```
$(vTotalCustomers)
```

What is displayed in the new column? Can you explain why?

19. Modify the expression as follows:

```
$(vNumCustomers) / $(vTotalCustomers)
```

20. Format the new expression as **Number** *(12.3%)* and add a label, *%Customers*.
21. Disable the edit mode and clear all the selections. What is the new column in the table representing? What is the percentage of customers in *Argentina*?



	Sales	Num Orders	Num Customers	%Customers
Argentina	$ 312562.22	94	3	3.3%
Austria	$ 727391.06	208	2	2.2%
Belgium	$ 220487.55	60	2	2.2%
Brazil	$ 1625584.66	461	9	9.8%

22. Select *Argentina* in the table. Looking at the new KPI you created, you can see there are 3 customers in this country. But why has the *%Customers* value changed?
23. Enable **Edit** mode in the sheet and open the **Variable** overview panel.
24. Modify the *vTotalCustomers* variable using the following **Definition**:

```
=count(distinct ALL CustomerID)
```

25. Disable edit mode and review the results in the table again. Make different selections in the *Country* field and see how the percentages remain correct.
26.

> Note: The **ALL** qualifier in the expression overwrites the user selections as it defines the whole data set. You can also use set analysis to achieve the same, as this is equivalent to **{1} TOTAL**. See more info about **TOTAL F** in the help site.